

Notes 6.5 - SOLVING SQUARE ROOT EQUATIONS

How to Solve Equations Involving Radicals:

3-step process:

(1) Isolate the radical. In other words, get it ALONE on one side of the equal sign.

(2) After the radical is ALONE, SQUARE BOTH SIDES

(3) Solve using algebra skills.

BUT watch out for this two situation!!!

- If the original problem has an x outside a radical, check for extraneous solutions!!
Check to make sure all possible answers make the ORIGINAL EQUATION true.

Examples. Solve for x .

1. $3 + \sqrt{2x - 3} = 8$

$$\begin{array}{r} -3 \quad -3 \\ \hline \sqrt{2x - 3} = 5 \end{array}$$

$$\begin{array}{r} 2x - 3 = 25 \\ +3 \quad +3 \end{array}$$

$$\frac{2x}{2} = \frac{28}{2}$$

$$\boxed{x = 14}$$

2. $4\sqrt{2x - 7} - 3 = 5$

$$\begin{array}{r} +3 \quad +3 \\ \hline \frac{4}{4} \sqrt{2x - 7} = \frac{8}{4} \end{array}$$

$$\sqrt{2x - 7} = 2$$

$$2x - 7 = 4$$

$$+7 \quad +7$$

$$\frac{2x}{2} = \frac{11}{2}$$

$$\text{or } \boxed{x = 5.5}$$

3. $\sqrt{x+20} = x$ ^{2 1/2} must check!

$$\begin{array}{r} x+20 = x^2 \\ -x \quad -20 \quad \quad -x \quad -20 \\ \hline \end{array}$$

$$0 = x^2 - x - 20$$

$$0 = (x-5)(x+4)$$

$$\begin{array}{r} x-5=0 \\ +5 \quad +5 \\ \hline x=5 \end{array}$$

$$\begin{array}{r} x+4=0 \\ -4 \quad -4 \\ \hline x=-4 \end{array}$$

check in $\sqrt{x+20} = x$

$x=5$: $\sqrt{5+20} = 5$?
 $\sqrt{25} = 5$?
 yay!

$x=-4$: $\sqrt{-4+20} = -4$?
 $\sqrt{16} = -4$?
 $4 = -4$

4. $\sqrt{5x-1} + 3 = x$ ² must check!

$$\sqrt{5x-1}^2 = (x-3)^2 \leftarrow \text{the whole side!}$$

$$5x-1 = (x-3)(x-3) \text{ F.O.I.L.}$$

$$5x-1 = x^2 - 3x - 3x + 9$$

$$\begin{array}{r} 5x-1 = x^2 - 6x + 9 \\ -5x+1 \quad \quad -5x+1 \\ \hline \end{array}$$

$$0 = x^2 - 11x + 10$$

$$0 = (x-1)(x-10)$$

$$\begin{array}{r} x-1=0 \\ +1 \quad +1 \\ \hline x=1 \end{array}$$

$$\begin{array}{r} x-10=0 \\ +10 \quad +10 \\ \hline x=10 \end{array}$$

check: $\sqrt{5x-1} + 3 = x$

$x=1$: $\sqrt{5(1)-1} + 3 = 1$?
 $\sqrt{4} + 3 = 1$
 $2+3 = 1$

$x=10$: $\sqrt{5(10)-1} + 3 = 10$?
 $\sqrt{49} + 3 = 10$
 $7+3 = 10$!!